FORMATIVE ASSESSMENT – 3 (FA-3)

MATHEMATICS • CLASS 9 • GENERATING NEW PROBLEMS

Problem 1

Express the linear equation **3x = 5y − 12** in the standard form **ax + by + c = 0**. Indicate the numerical values of **a**, **b**, and **c**, and determine two distinct solutions for this equation.

Problem 2

The cost of a notebook is twice the cost of a pen. Write a linear equation in two variables to represent this statement. If the total cost of 3 notebooks and 4 pens is ₹100, find the individual cost of a notebook and a pen.

Problem 3

Find the value of **k** if the point **(2, −3)** lies on the graph of the linear equation **4x + ky = 17**. Formulate a similar problem by changing the coordinates and constant.

Problem 4

In an isosceles triangle **ABC** with **AB = AC**, the bisectors of ∠**B** and ∠**C** intersect at point **O**. Show that **OB = OC** and that line segment **AO** bisects ∠**A**.

Problem 5

Two triangles **PQR** and **XYZ** have **PQ = XY** and **QR = YZ**. What additional condition is required to establish congruence between the two triangles using the **SAS** criterion? What condition is needed for the **SSS** criterion?